Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "electric field" and "charge" to explain your answer. Sentence starters are optional.

electric field: force per positive test charge at a location

charge: a property responsible for electric interactions

vector: a quantity with size and direction

01 Warm-up: bring back an earlier idea

In separate cases, an ideal blackbody temperature becomes **3** times as large and a sound frequency becomes **5** times as large at fixed sound speed. **Compare the two wavelength changes**.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=**4 N/C** right, a **+2 C** test charge feels **8 N** right. A-**2 C** charge feels **8 N** left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Field / first contribution

Force / second contribution

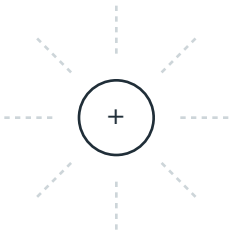
Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A tiny charged bead has charge **-2.0 × 10⁻⁶ C** in a uniform electric field of **300 N/C** *east*.
1. **Calculate its electric force**.
2. Use separate arrows to **explain why force** and field point in different directions.

Word help: uniform = constant throughout the stated interval

Sentence starter: Both … . They differ in … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

05 Explain your reasoning

If you remove the test charge, does the source charge's field disappear? **Explain the role of a** test charge.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A closed-open pipe has length **0.4 m**. Use λ=**4 L** for the fundamental and v=**340 m/s**. **Find its fundamental frequency**.
Sketch the displacement pattern and **label the pipe length**.

Sentence starter: I found … [unit]. This value means … in this situation.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3

07 Exit check

At twice the distance from a point charge, how does field magnitude change? Say what the calculation treats as true.

Word help: magnitude = size of a quantity without its direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **-3** microcoulomb charge feels a **0.012 N** force upward. **Find the electric-field vector**.

Sentence starter: I found … [unit]. This value means … in this situation.

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3